

RF and Microwave Engineering Numerical Companion

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · Chapter 4, all 8 PYQ waveguide and junction problems, 2069–2082 BS

4. RF/Microwave Components and Devices: Numerical Problems

17 solved PYQ problems · **≈127 marks** across 17 papers · 5 waveguide sets, 3 tee derivations, 4 junction networks, 3 identifications, 2 radar designs

READ THIS FIRST

Chapter 4 numerals come in two kinds, and neither needs more than a calculator's square root.

- **Waveguide arithmetic** (Problems 1 to 5). Everything hangs off one number, the **factor** $F = \sqrt{1 - (f_c/f)^2}$. Compute it once, write it in a box, and every other quantity is a multiplication or a division by it:

$$\lambda_g = \frac{\lambda}{F}, \quad v_p = \frac{c}{F}, \quad v_g = cF, \quad Z_{TE} = \frac{\eta_0}{F}, \quad Z_{TM} = \eta_0 F.$$

Always start by naming the dominant mode and checking $f > f_c$; a guide below cut-off has no λ_g or v_p to find.

- **Tee matrices** (Problems 6 to 17). Build each tee's matrix from its physics (M2 below), then **follow one unit wave through**: out of one tee, into the next, out again. A matched termination deletes a row and column; a short reflects with $\Gamma = -1$; a connection hands a leaving wave to the other tee as an arriving one. Column power sums are the self-check, and the four tests are Ch3 companion M1.
- **Watch the normalising factor**. Papers print magic-tee matrices without the $1/\sqrt{2}$. Spotting that it is missing is worth a mark on its own.
- **Sign, not port number, names the arm**. A column whose entries flip sign is the E-arm (difference); a column whose entries match is the H-arm (sum). Papers swap which port number carries which.
- Constants used by the papers and here: $c = 3 \times 10^8$ m/s, $\eta_0 = 120\pi = 376.99 \Omega$. The guide answers are asserted in `src/wg.py`, the tee matrices in `src/sparm.py`.

M. Three Matrix Methods for the Tee Problems

M2: build a device matrix from its physics (Problems 6, 7, 8)

1. Write the blank $n \times n$ matrix.
2. Impose the **phase facts**: which arms come out in phase ($S_{2k} = +S_{1k}$) and which in antiphase ($S_{2k} = -S_{1k}$).
3. Impose the **matching facts**: $S_{kk} = 0$ for every port the question calls “perfectly matched to the junction”.
4. Impose **reciprocity**, $S_{ij} = S_{ji}$, and count what is left. Both a plane tee and a magic tee leave exactly **4 unknowns**.
5. Apply the **unitary property** $[S][S]^* = [I]$ and read off the equations row by row. Each row-column product gives one equation; solve.

M3: reduce an n -port by terminating ports (Problems 11, 12)

1. A termination on port k forces $a_k = \Gamma_k b_k$, with $\Gamma = -1$ for a **short**, $+1$ for an **open**, 0 for a **matched load**, all measured at the junction reference plane.
2. Write out $b_i = \sum_j S_{ij} a_j$ for every port.
3. Substitute the constraint for each terminated port and eliminate its a and b .
4. What is left is the smaller matrix seen from the remaining ports.

M4: power bookkeeping (Problems 16, 17)

1. Write the drive vector $[a]$ that the connection creates. Two equal in-phase transmitters on ports 1 and 2 is $[a] = [a, a, 0, 0]^T$.
2. Multiply: $[b] = [S][a]$.
3. Power leaving port i is $|b_i|^2$, in the same units as $|a|^2$.
4. Check the books: $\sum_i |b_i|^2 \leq \sum_j |a_j|^2$, with equality for a lossless network. $(1/\sqrt{2})^2 = 0.5$: half the power.

Problem 1: Full parameter set of an air-filled rectangular guide 6

80 Ba · width 3 cm, 6 GHz, dominant mode

QUESTION AS PRINTED 80 Ba [6+4]

For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler.

Given

- Air filling: $v = c = 3 \times 10^8$ m/s, $\eta = \eta_0 = 120\pi \Omega$.
- Width (broad wall) $a = 3$ cm = 0.03 m; $f = 6$ GHz.
- Dominant mode of a rectangular guide: TE_{10} ($m = 1, n = 0$), so b is not needed.

Step 1: cut-off frequency and wavelength of TE_{10}

- $f_c = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + 0} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.03} = \boxed{5 \text{ GHz}}$.
- $\lambda_c = 2a = \boxed{6 \text{ cm}}$ (check: $c/f_c = 3 \times 10^8 / 5 \times 10^9 = 0.06$ m).
- $f = 6 \text{ GHz} > f_c = 5 \text{ GHz} \Rightarrow$ **the guide propagates**, so the remaining quantities exist.

Step 2: the factor, computed once

- $f_c/f = 5/6 = 0.83333$; $(f_c/f)^2 = 0.69444$; $1 - 0.69444 = 0.30556$.
- $F = \sqrt{0.30556} = \boxed{0.55277}$ (exactly $\sqrt{11/6}$).
- Free-space wavelength $\lambda = c/f = 3 \times 10^8 / 6 \times 10^9 = 5$ cm.

Step 3: every other quantity from F

Quantity	Formula	Substitution	Value
Group velocity	$v_g = cF$	$3 \times 10^8 \times 0.55277$	1.658 $\times 10^8$ m/s
Phase velocity	$v_p = c/F$	$3 \times 10^8 / 0.55277$	5.427 $\times 10^8$ m/s
Guide wavelength	$\lambda_g = \lambda/F$	$5/0.55277$	9.045 cm
Impedance (TE)	$Z_{TE} = \eta_0/F$	$376.99/0.55277$	682.0 Ω
Phase constant	$\beta = 2\pi/\lambda_g$	$2\pi/0.090453$	69.46 rad/m

- **Checks:** $v_p v_g = 5.427 \times 10^8 \times 1.658 \times 10^8 = 9.0 \times 10^{16} = c^2 \checkmark$; and $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$: $400 = 122.2 + 277.8 \text{ m}^{-2} \checkmark$.
- “Characteristic impedance” of a TE guide is its wave impedance Z_{TE} ; it is above η_0 , as it must be for TE.
- **Rider (4 marks):** coupling, directivity, isolation, insertion loss, with $I = C + D$. See theory §4.8.

Problem 2: Same parameters, with suitably assumed dimensions 6

77 Ch · the choice of dimensions is part of the marks

QUESTION AS PRINTED 77 Ch [6]

For a rectangular waveguide, with suitably assumed breadth, width and frequency, (for dominant mode), determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance.

Step 0: choose the data, and justify it

- Take the standard X-band guide **WR-90**: $a = 2.286$ cm, $b = 1.016$ cm, air filled, at $f = 10$ GHz.
- Why it is a good choice: it runs in **single-mode** TE_{10} at 10 GHz. The next modes are cut off:
 - TE_{20} : $f_c = c/a = 3 \times 10^8 / 0.02286 = 13.12 \text{ GHz} > 10 \text{ GHz}$.
 - TE_{01} : $f_c = c/2b = 3 \times 10^8 / 0.02032 = 14.76 \text{ GHz} > 10 \text{ GHz}$.
- Any other data is acceptable if it is stated and above cut-off; the method does not change.

Step 1: cut-off

- $f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.02286} = \boxed{6.562 \text{ GHz}}$, $\lambda_c = 2a = 4.572$ cm. $10 > 6.562 \checkmark$ propagates.

Step 2: the factor

- $f_c/f = 0.65617$; squared 0.43056; $1 - 0.43056 = 0.56944$; $F = \sqrt{0.56944} = \boxed{0.75461}$.
- $\lambda = c/f = 3 \text{ cm}$.

Step 3: the asked quantities

Quantity	Formula	Substitution	Value
Phase velocity	$v_p = c/F$	$3 \times 10^8 / 0.75461$	$3.976 \times 10^8 \text{ m/s}$
Guide wavelength	$\lambda_g = \lambda/F$	$3 / 0.75461$	3.976 cm
Impedance (TE)	$Z_{TE} = \eta_0/F$	$376.99 / 0.75461$	499.6 Ω
Group velocity (extra)	$v_g = cF$	$3 \times 10^8 \times 0.75461$	$2.264 \times 10^8 \text{ m/s}$

- Check: $1/\lambda^2 = 1111.1$ and $1/\lambda_g^2 + 1/\lambda_c^2 = 632.7 + 478.4 = 1111.1 \text{ m}^{-2} \checkmark$.

Problem 3: Does the guide pass the signal? Cut-off checks 3 2

70 Bh, 80 Bh · one method, two instances; 70 Bh worked in full

QUESTION AS PRINTED 70 Bh [7+3]

Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in TE₁₀ made in a rectangular waveguide with a wall separation of 5 cm.

QUESTION AS PRINTED 80 Bh [1+1+2]

What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz?

Method

1. Name the mode (TE₁₀, the dominant one) and read the broad-wall width a .
2. $f_c = c/2a$.
3. Compare: $f > f_c$ propagates; $f < f_c$ is evanescent.
4. For "show that", add the attenuation: $\gamma = \alpha = \sqrt{k_c^2 - k^2}$ is real.

70 Bh worked: $a = 5 \text{ cm}$, $f = 1 \text{ GHz}$

- $f_c = \frac{3 \times 10^8}{2 \times 0.05} = 3 \times 10^9 \text{ Hz} = \boxed{3 \text{ GHz}}$.
- $f = 1 \text{ GHz} < f_c = 3 \text{ GHz} \Rightarrow$ below cut-off.
- Confirm with γ :
 - $k_c = \pi/a = \pi/0.05 = 62.832 \text{ rad/m}$, $k_c^2 = 3947.8$.
 - $k = 2\pi f/c = 2\pi \times 10^9 / 3 \times 10^8 = 20.944 \text{ rad/m}$, $k^2 = 438.6$.
 - $\gamma^2 = k_c^2 - k^2 = 3947.8 - 438.6 = 3509.2 > 0 \Rightarrow \gamma$ is real: $\gamma = \alpha = \sqrt{3509.2} = 59.24 \text{ Np/m}$, $\beta = 0$.
- The field decays as $e^{-59.24z}$; to $1/e$ in $1/59.24 = 1.69 \text{ cm}$, i.e. $20 \log_{10} e \times 59.24 = 514.5 \text{ dB/m}$. **No wave travels: the 1 GHz signal cannot propagate.**

Both instances

Paper	Given	$f_c = c/2a$	Compare	Answer
70 Bh	$a = 5 \text{ cm}$, 1 GHz	3.00 GHz	$1 < 3$	cannot propagate ; $\alpha = 59.2 \text{ Np/m}$
80 Bh	$a = 2.254 \text{ cm}$, 6 GHz	$3 \times 10^8 / 0.04508 = 6.65 \text{ GHz}$	$6 < 6.65$	not supported

- **80 Bh theory part:** dominant mode = lowest cut-off mode; rectangular TE₁₀, circular TE₁₁ (§4.2).
- The handwritten local example (Waveguide.pdf p6) runs the same 2.254 cm guide at 1 GHz and reaches the same verdict with $f_c = 6.65 \text{ GHz}$.

Problem 4: Ranking TM modes when $a = 3b$ 2+2

72 Ma

QUESTION AS PRINTED 72 Ma [6+2+2]

Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of $a = 3b$ and find the dominant mode among TM_{01} , TM_{10} , TM_{11} , TM_{21} , TM_{12} , TM_{02} and TM_{20} .

Step 1: put the cut-off formula in terms of a only

$$- f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \text{ and } b = a/3 \Rightarrow n/b = 3n/a:$$

$$f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$$

- So each mode is ranked by the number $\sqrt{m^2 + 9n^2}$ (in units of $c/2a$).

Step 2: test every listed mode

Mode	(m, n)	Exists? (TM needs $m, n \geq 1$)	$\sqrt{m^2 + 9n^2}$	Rank
TM_{01}	(0, 1)	no, $E_z \propto \sin 0 = 0$	n/a	excluded
TM_{10}	(1, 0)	no	n/a	excluded
TM_{11}	(1, 1)	yes	$\sqrt{10} = 3.162$	1, dominant
TM_{21}	(2, 1)	yes	$\sqrt{13} = 3.606$	2
TM_{12}	(1, 2)	yes	$\sqrt{37} = 6.083$	3
TM_{02}	(0, 2)	no	n/a	excluded
TM_{20}	(2, 0)	no	n/a	excluded

- **Answer:** among the listed modes the dominant one is TM_{11} , $f_c = 3.162 c/2a$.
- Worth one line: the guide's overall dominant mode is TE_{10} at $1 \times c/2a$, which the list does not include.
- Riders: magic tee §4.5; dominant vs degenerate §4.2.

Problem 5: Dominant TE and TM modes of a circular guide, with numbers 2

75 Bh · plus the local worked example as practice

QUESTION AS PRINTED 75 Bh [8+2]

Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes.

Step 1: cut-off in terms of the roots

- TE_{nm} : $f_c = \frac{p'_{nm} c}{2\pi a}$; TM_{nm} : $f_c = \frac{p_{nm} c}{2\pi a}$.
- Same a and c for every mode, so **the smallest root is the dominant mode.**

Step 2: compare the smallest roots

Family	Smallest root	Mode	Verdict
TE	$p'_{11} = 1.841$ (first zero of J'_1)	TE_{11}	dominant TE mode
TE	$p'_{01} = 3.832$ (first zero of J'_0)	TE_{01}	higher
TM	$p_{01} = 2.405$ (first zero of J_0)	TM_{01}	dominant TM mode
TM	$p_{11} = 3.832$ (first zero of J_1)	TM_{11}	higher

- $1.841 < 2.405 \Rightarrow TE_{11}$ **is the dominant mode of the guide**, TM_{01} the dominant TM mode. The derivation part is theory §4.3.

Practice (Waveguide.pdf p13): air-filled, radius 2 cm

Mode	Substitution	f_c	$\lambda_c = 2\pi a/p$
TE_{11}	$1.8412 \times 3 \times 10^8 / (2\pi \times 0.02)$	4.40 GHz	6.83 cm
TM_{01}	$2.4048 \times 3 \times 10^8 / (2\pi \times 0.02)$	5.74 GHz	5.23 cm

- Does 3 GHz propagate? $3 < 4.40 \text{ GHz} \Rightarrow$ **no**, not even in the dominant mode.

Problem 6: Derive the S-matrix of an E-plane Tee 4 8

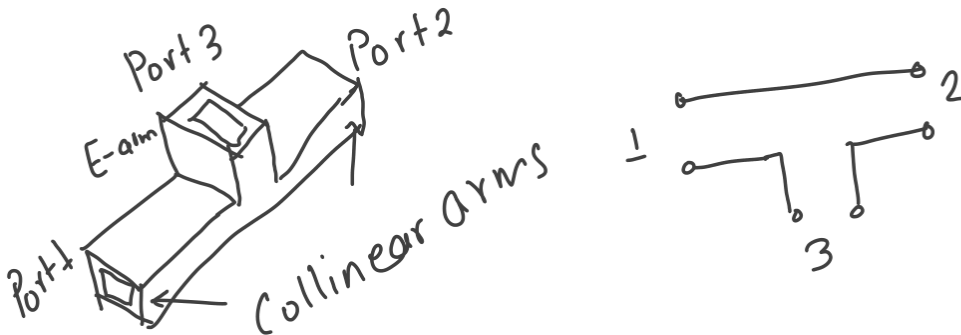
80 Bh, 81 Ba · the model 3-port derivation; learn this and Problems 7 and 8 are the same script

QUESTION AS PRINTED 80 Bh [4+4]

Briefly explain the need of S-parameters and their properties in microwave analysis. Explain three-port network parameters using an example of an E-plane Tee.

Given

- An E-plane (series) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** side arm, assumed **perfectly matched to the junction**.
- The junction is a passive, lossless, isotropic waveguide structure.



E-plane tee: 3D view and schematic, ports labelled as used below. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: write the blank matrix

- A 3-port has $3^2 = 9$ coefficients:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \quad (\text{i})$$

Step 2: impose the three physical facts

Physical fact	Equation	Why
An input at the E-arm leaves ports 1 and 2 180° out of phase	$S_{23} = -S_{13}$ (ii)	The E-field of the side arm drives the two collinear arms in opposition
Port 3 is perfectly matched to the junction	$S_{33} = 0$ (iii)	Nothing incident on the E-arm is reflected
The junction is passive and isotropic	$S_{ij} = S_{ji}$ (iv)	Reciprocity: $S_{12} = S_{21}, S_{23} = S_{32}, S_{31} = S_{13}$

- Substituting (ii), (iii) and (iv) into (i):

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \quad (\text{v})$$

- **Four unknowns remain:** $S_{11}, S_{12}, S_{22}, S_{13}$. Four equations are needed, and the unitary property supplies them.

Step 3: apply the unitary (lossless) property

- A lossless network satisfies $[S][S]^*{}^T = [I]$, and since (v) is symmetric this is simply $[S][S]^* = [I]$:

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 4: read off one equation per row-column product

Product	Equation	What it gives	
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1 \Rightarrow 2 S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$	(vi)
$R_3 \cdot C_1$	$S_{13}S_{11}^* - S_{13}S_{12}^* + 0 = 0 \Rightarrow S_{11}^* = S_{12}^*$	$S_{11} = S_{12}$	(vii)
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + S_{13} ^2 = 1$, and with (vi) and (vii) $2 S_{11} ^2 = 1 - \frac{1}{2}$	$ S_{11} ^2 = \frac{1}{4}$, so $S_{11} = S_{12} = \frac{1}{2}$	
$R_2 \cdot C_2$	$ S_{12} ^2 + S_{22} ^2 + S_{13} ^2 = 1$, i.e. $\frac{1}{4} + S_{22} ^2 + \frac{1}{2} = 1$	$ S_{22} ^2 = \frac{1}{4}$, so $S_{22} = \frac{1}{2}$	

Answer

- Substituting all four values back into (v):

$$[S]_{\text{E-tee}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Reading the answer back (this is the “explain” half of the marks)

- **Power split:** $|S_{13}|^2 = |S_{23}|^2 = \frac{1}{2}$, so an input at the E-arm divides **equally** between the collinear arms.
- **Phase:** $S_{23} = -S_{13}$, so those two outputs are **antiphase**; this is what makes the E-tee a **difference** junction.
- **Matching:** $S_{33} = 0$ but $S_{11} = S_{22} = \frac{1}{2}$, so the collinear arms are **not matched**: a quarter of the power incident on port 1 comes straight back. That is the 3-port theorem of §3.5 appearing as arithmetic.
- **Isolation:** $S_{12} = \frac{1}{2} \neq 0$, so the collinear arms are **not isolated** from each other either. Adding the H-arm to fix both defects is exactly what turns a tee into a magic tee (Problem 8).
- **Check:** every column sums to $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1 \checkmark$ lossless.

Problem 7: Derive the S-matrix of an H-plane Tee 8

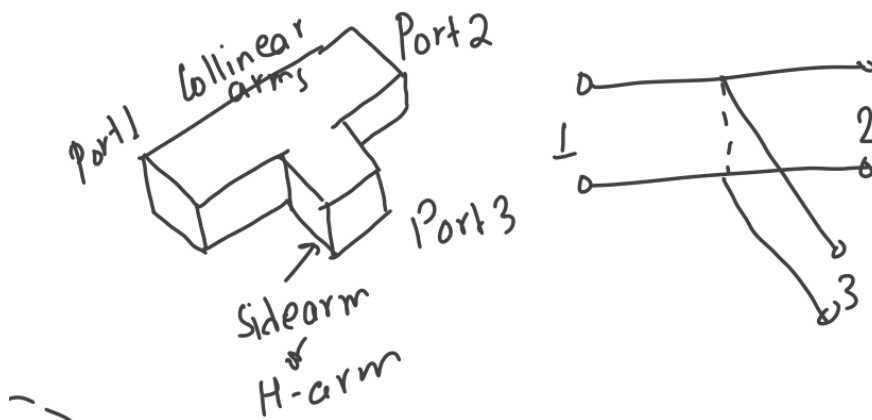
81 Ba · the same script as Problem 6 with one sign changed

QUESTION AS PRINTED 81 Ba [8]

Prepare S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses.

Given

- An H-plane (shunt) tee: ports 1 and 2 collinear, port 3 the **H-arm** side arm, perfectly matched to the junction.



H-plane tee: the side arm lies in the plane of the magnetic field. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: the one fact that differs from Problem 6

- In an H-plane tee an input at the side arm reaches the two collinear arms **in phase**:

$$S_{13} = S_{23} \quad (\text{ii}')$$

- Everything else is unchanged: $S_{33} = 0$ (iii) and $S_{ij} = S_{ji}$ (iv).
- So the working matrix is

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \quad (\text{v}')$$

with the same four unknowns.

Step 2: unitary property, row by row

Product	Equation	What it gives
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_3 \cdot C_1$	$S_{13}S_{11}^* + S_{13}S_{12}^* = 0 \Rightarrow S_{11}^* = -S_{12}^*$	$S_{11} = -S_{12}$
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + \frac{1}{2} = 1 \Rightarrow 2 S_{11} ^2 = \frac{1}{2}$	$S_{11} = \frac{1}{2}$, so $S_{12} = -\frac{1}{2}$
$R_2 \cdot C_2$	$\frac{1}{4} + S_{22} ^2 + \frac{1}{2} = 1$	$S_{22} = \frac{1}{2}$

- **The sign moved.** In the E-tee the minus sign sat in the side-arm column; here it sits in S_{12} instead. That single difference is the whole distinction between the two tees.

Answer, and the duplexer the same question asks for

- H-plane tee:

$$[S]_{\text{H-tee}} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- **Duplexer**, taken as an ideal 3-port ferrite circulator with port 1 the transmitter, port 2 the antenna and port 3 the receiver:

$$[S]_{\text{duplexer}} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Matched at every port ($S_{ii} = 0$), lossless (every column sums to 1) and **deliberately non-reciprocal** ($S_{21} = 1$ but $S_{12} = 0$), which is the only way a 3-port can be both matched and lossless (§3.5).

Uses, which the question asks for explicitly

Device	Behaviour	Uses
E-plane tee	Equal split, outputs in antiphase; difference junction	Balanced mixers, phase comparators, the difference arm of a magic tee, feeding balanced antennas
H-plane tee	Equal split, outputs in phase; sum junction	In-phase power dividers and combiners, feeding two antennas from one source, the sum arm of a magic tee
Duplexer	One antenna shared by transmitter and receiver, with the receiver isolated from the transmitter	Radar, and any single-antenna transceiver; protects the sensitive receiver front end from the transmit pulse

Problem 8: Derive the S-matrix of a Magic Tee 3+3 8 10

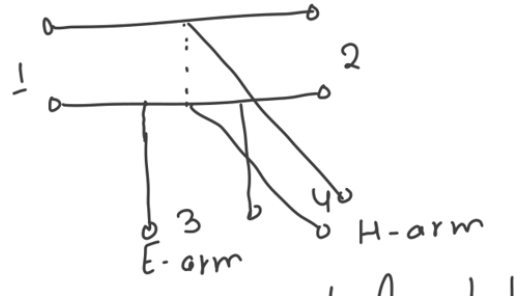
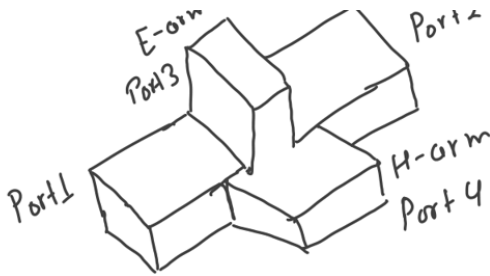
72 Ash, 74 Bh, 75 Bh, 78 Ch, 81 Bh · the single highest-value derivation in the chapter

QUESTION AS PRINTED The device behind five identification questions

Describe the operational principles of a magic tee based on its S-parameters, and derive its scattering matrix.

Given

- A hybrid (magic) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** and port 4 is the **H-arm**. Both side arms are perfectly matched to the junction.



Magic tee: an E-plane tee and an H-plane tee sharing the same junction. Source: RF Pulchowk, S-parameter derivation sheets.

Step 1: write the blank 4×4

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \quad (\text{i})$$

Step 2: impose the five physical facts

Physical fact	Equation	Why
It contains an E-plane tee, so an input at the E-arm (port 3) leaves ports 1 and 2 antiphase	$S_{23} = -S_{13}$	(ii) Same argument as Problem 6
It contains an H-plane tee, so an input at the H-arm (port 4) leaves ports 1 and 2 in phase	$S_{24} = +S_{14}$	(iii) Same argument as Problem 7
An input at the E-arm produces no output at the H-arm, and the reverse	$S_{34} = S_{43} = 0$	(iv) The two side arms are geometrically orthogonal; the E-arm excites an antisymmetric field, the H-arm a symmetric one
Both side arms are perfectly matched to the junction	$S_{33} = S_{44} = 0$	(v) Stated in the question
No output at port 2 for an input at port 1, and the reverse	$S_{12} = S_{21} = 0$	(vi) The collinear arms are decoupled, which is the “magic”

- With reciprocity $S_{ij} = S_{ji}$ (vii), equations (ii) to (vii) reduce (i) to

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix} \quad (\text{viii})$$

- Four unknowns:** $S_{11}, S_{22}, S_{13}, S_{14}$.

Step 3: unitary property, row by row

Product	Equation	What it gives
$R_3 \cdot C_3$	$ S_{13} ^2 + -S_{13} ^2 + 0 + 0 = 1$, i.e. $2 S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_4 \cdot C_4$	$ S_{14} ^2 + S_{14} ^2 + 0 + 0 = 1$, i.e. $2 S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$
$R_1 \cdot C_1$	$ S_{11} ^2 + 0 + S_{13} ^2 + S_{14} ^2 = 1$, i.e. $ S_{11} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$ S_{11} ^2 = 0$, so $\mathbf{S}_{11} = \mathbf{0}$
$R_2 \cdot C_2$	$0 + S_{22} ^2 + S_{13} ^2 + S_{14} ^2 = 1$	$\mathbf{S}_{22} = \mathbf{0}$

- **This is the striking result:** matching the two side arms and decoupling the collinear pair **forces** the collinear arms to be matched as well. Nothing was assumed about S_{11} or S_{22} ; the unitary condition produced them.

Answer

$$[S]_{\text{magic tee}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Operational principles, read straight off the matrix

Drive	Outputs	Consequence
E-arm (port 3)	$b_1 = \frac{1}{\sqrt{2}}a_3, b_2 = -\frac{1}{\sqrt{2}}a_3,$ $b_4 = 0$	Equal split, antiphase, nothing into the H-arm
H-arm (port 4)	$b_1 = b_2 = \frac{1}{\sqrt{2}}a_4, b_3 = 0$	Equal split, in phase, nothing into the E-arm
Port 1	$b_3 = \frac{1}{\sqrt{2}}a_1, b_4 = \frac{1}{\sqrt{2}}a_1,$ $b_2 = 0$	Splits into both side arms, nothing reaches port 2
Ports 1 and 2 together	$b_4 \propto (a_1 + a_2),$ $b_3 \propto (a_1 - a_2)$	H-arm is the sum arm, E-arm the difference arm

- **Why “magic”:** all four ports are matched *and* two pairs (1–2 and 3–4) are completely isolated, which a 3-port can never achieve.
- **Uses:** power combining and splitting (§4.7), balanced mixers, duplexers, impedance bridges, monopulse radar comparators.
- **In practice** the ideal matrix needs matching posts or irises at the junction; without them S_{11} and S_{22} are not zero.

Problem 9: Two magic tees joined at their E-arms 10

77 Ch · derived by connecting two 4-port matrices; asserted in `wg.py`

QUESTION AS PRINTED 77 Ch [10]

Explain the properties of two Magic Tees if one connects their E-arms and derive its S-matrix.

Given: the two tees

- Each tee, ports (1, 2) collinear, 3 = E-arm, 4 = H-arm, $r = 1/\sqrt{2}$ (§4.5): $[S_T] = r \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$.
- Tee A ports 1, 2, 3, 4; tee B ports 1', 2', 3', 4'. The E-arms 3 and 3' are joined, so they disappear as external ports.
- Joining means: the wave **leaving** arm 3 **enters** arm 3', and vice versa: $a_{3'} = b_3, a_3 = b_{3'}$.
- External ports left: 1, 2, 4 (tee A) and 1', 2', 4' (tee B) $\Rightarrow$ a **6-port**, 6×6 matrix.

Step 1: drive port 1 with a unit wave ($a_1 = 1$, all other inputs zero)

- Inside tee A: $b_3 = S_{31} = r, b_4 = S_{41} = r, b_2 = S_{21} = 0$.
- $b_3 = r$ enters tee B's E-arm: $a_{3'} = r$.
- Inside tee B: $b_{1'} = S_{1'3'} a_{3'} = r \cdot r = \frac{1}{2}, b_{2'} = S_{2'3'} a_{3'} = -r \cdot r = -\frac{1}{2}, b_{4'} = S_{4'3'} a_{3'} = 0$.
- Nothing reflects back into arm 3, because $S_{3'3'} = 0$ (matched E-arm).

- Column for port 1: $(b_1, b_2, b_4, b_{1'}, b_{2'}, b_{4'}) = (0, 0, r, \frac{1}{2}, -\frac{1}{2}, 0)$.
- Power check: $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \checkmark$.

Step 2: the other drives, same way

- Port 2: $b_3 = S_{32} = -r, b_4 = r$; tee B receives $-r \Rightarrow b_{1'} = -\frac{1}{2}, b_{2'} = +\frac{1}{2}$.
- Port 4 (H-arm A): $b_1 = b_2 = r$, and $b_3 = S_{34} = 0 \Rightarrow$ **nothing crosses** to tee B.
- Ports 1', 2', 4': mirror images, by symmetry of the connection.

Step 3: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 1' & 2' & 4' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \\ 1' \\ 2' \\ 4' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & r & -\frac{1}{2} & \frac{1}{2} & 0 \\ r & r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & r \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ 0 & 0 & 0 & r & r & 0 \end{bmatrix} \end{matrix}$$

$r = 1/\sqrt{2}$. Identical to the `connect()` result in `wg.py`.

Step 4: properties

- **Matched**: all six diagonal entries are 0.
- **Reciprocal**: $[S] = [S]^T$.
- **Lossless**: every column's powers sum to 1.
- **Collinear isolation kept**: $S_{12} = S_{1'2'} = 0$.
- **H-arms isolated** from each other and from the far tee: $S_{44'} = 0$.
- **Difference channel crosses**: an antiphase pair at (1, 2) appears entirely at (1', 2'), again antiphase.
- **Sum channel stays**: an in-phase pair at (1, 2) leaves entirely by 4.
- Use: a sum/difference separator, e.g. two monopulse channels feeding a common difference path.

Problem 10: Two magic tees connected in the H-plane 5

78 Ch · a short note; the Problem 9 method with the H-arms joined

QUESTION AS PRINTED 78 Ch [5]

Write short notes on: Combinational effect of two magic tees connected in H-plane.

The paper's typos ("Combinational", "magic tees") are reproduced as printed; read "combinational effect of two magic tees".

Step 1: join H-arm 4 to H-arm 4' ($a_{4'} = b_4, a_4 = b_{4'}$) and follow port 1

- Tee A: $b_3 = S_{31} = r, b_4 = S_{41} = r$.
- $b_4 = r$ enters tee B's H-arm: $b_{1'} = S_{14} r = \frac{1}{2}, b_{2'} = S_{24} r = +\frac{1}{2}$ (in phase), $b_{3'} = S_{34} r = 0$.
- From port 2: $b_4 = S_{42} = +r$ as well, so tee B again gets $+\frac{1}{2}, +\frac{1}{2}$.

Step 2: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 1' & 2' & 3' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 1' \\ 2' \\ 3' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -r & \frac{1}{2} & \frac{1}{2} & 0 \\ r & -r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -r \\ 0 & 0 & 0 & r & -r & 0 \end{bmatrix} \end{matrix}$$

Short-note points

- Matched at all six ports, reciprocal, lossless.
- The **sum** (in-phase) signal crosses between the tees through the joined H-arms and re-appears **in phase** at 1', 2'.
- The **difference** signal stays in its own tee's E-arm.
- Collinear ports of each tee stay isolated; the two E-arms are isolated.
- The exact mirror of Problem 9: E-joining passes the difference, H-joining passes the sum.

Problem 11: Hybrid tee with a terminated arm 4+4+2

82 Ba · 70 Ma's "hybrid tee" [8] is answered by §4.5 plus this

QUESTION AS PRINTED 82 Ba [4+4+2]

Using proper S-Matrices, mention the properties of a Hybrid-Tee in the situation of (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planner arms.

Step 0: what “terminated” does to a matrix

- Terminated = a **matched load**, $\Gamma = 0$: whatever leaves that arm is absorbed, nothing comes back, so $a_k = 0$.
- Therefore row k and column k simply **drop out**; the rest of the matrix is unchanged.
- Start from the magic tee, ports 1, 2 collinear (the co-planar arms), 3 = E, 4 = H.
- If the paper meant a **short**, $\Gamma = -1$, the answer is different; that is the 82 Bh case, Problem 12.

Case	Reduced S-matrix	Properties
(a) E-arm terminated, ports (1, 2, 4)	$\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & r \\ r & r & 0 \end{bmatrix}$	Matched and reciprocal. From the H-arm: equal, in-phase halves at 1 and 2, no loss. Ports 1 and 2 isolated . From port 1: only half reaches port 4, the other half is absorbed in the E-arm load, so the 3-port is not lossless . An in-phase power divider.
(b) H-arm terminated, ports (1, 2, 3)	$\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & -r \\ r & -r & 0 \end{bmatrix}$	Matched and reciprocal. From the E-arm: equal, antiphase halves at 1 and 2. Ports 1 and 2 isolated. From port 1: half to port 3, half lost in the H-arm load. An antiphase (180°) power divider.
(c) Both co-planar arms terminated, ports (3, 4)	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	E and H arms both matched and completely isolated . Any power entering either side arm splits into the two loads and nothing comes out anywhere: the tee acts as a matched termination for both.

- Why (a) and (b) **must** be lossy: they are matched, reciprocal 3-ports, and §3.5 proves no such 3-port can be lossless. The load is where the missing half goes.
- Column check, (a): column 1 = $0 + 0 + \frac{1}{2} = \frac{1}{2}$ (half absorbed); column 4 = $\frac{1}{2} + \frac{1}{2} = 1$ (lossless from the H-arm).

Problem 12: Magic tee with both E- and H-arms short-circuited 4+4

82 Bh · verified/added no source note carries this; derived here and asserted in `sparam.py`

QUESTION AS PRINTED 82 Bh [4+4]

Design a model of a three-port network and define its characteristic parameters. Prepare the S-matrix of a rectangular magic tee having both E- and H-arms shorted, and explain how it behaves.

Given

- The ideal magic tee of Problem 8, with ports 3 (E-arm) and 4 (H-arm) **short-circuited at the junction reference plane**.
- A short reflects with $\Gamma = -1$, so $a_3 = -b_3$ and $a_4 = -b_4$.

Step 1: write out the four scattering equations

- From $[b] = [S][a]$ with the magic-tee matrix:

$$\begin{aligned} b_1 &= \frac{1}{\sqrt{2}}(a_3 + a_4), & b_2 &= \frac{1}{\sqrt{2}}(-a_3 + a_4) \\ b_3 &= \frac{1}{\sqrt{2}}(a_1 - a_2), & b_4 &= \frac{1}{\sqrt{2}}(a_1 + a_2) \end{aligned}$$

Step 2: substitute the short-circuit conditions

- $a_3 = -b_3 = -\frac{1}{\sqrt{2}}(a_1 - a_2)$.
- $a_4 = -b_4 = -\frac{1}{\sqrt{2}}(a_1 + a_2)$.

Step 3: eliminate and collect

- Into the expression for b_1 :

$$b_1 = \frac{1}{\sqrt{2}} \left[-\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = -\frac{1}{2} [(a_1 - a_2) + (a_1 + a_2)] = -\frac{1}{2}(2a_1) = -a_1$$

- Into the expression for b_2 :

$$b_2 = \frac{1}{\sqrt{2}} \left[+\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = \frac{1}{2} [(a_1 - a_2) - (a_1 + a_2)] = \frac{1}{2}(-2a_2) = -a_2$$

- The a_2 terms cancel in b_1 and the a_1 terms cancel in b_2 : the two reflections arriving from the shorted arms add in the driven arm and subtract in the other one.

Answer

- Seen from the two collinear arms, the shorted magic tee is the two-port

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

How it behaves

- **Total reflection:** $|S_{11}| = |S_{22}| = 1$, so every watt fed into port 1 comes back out of port 1. Return loss = 0 dB, VSWR = ∞ .
- **Phase reversal:** the -1 is a 180° shift, exactly what an ideal short does. The tee behaves as a **pair of independent short circuits**, one at each collinear arm.
- **Complete isolation survives:** $S_{12} = S_{21} = 0$, so the two arms still cannot see each other even though both are totally reflecting.
- **Still lossless:** each column sums to $|-1|^2 = 1$; nothing is dissipated, everything is returned.
- **Still reciprocal:** $[S] = [S]^T$.
- **Use:** a reflective phase shifter or a calibration short. Moving the shorts a distance ℓ out from the junction changes Γ from -1 to $-e^{-2j\beta\ell}$, and the same algebra then gives $S_{11} = S_{22} = -e^{-2j\beta\ell}$ with S_{12} still zero, so a tunable phase with a constant unity magnitude.

Problem 13: Identify the device from a numeric 4-port matrix 10

78 Ch · the printed matrix is missing its normalising factor

QUESTION AS PRINTED 78 Ch [10]

Identify and explain the properties of a microwave passive device having the following S-matrix, with necessary diagram. $[S] =$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Given

- A 4×4 matrix, so the device has **four ports**.
- **The paper prints the first row as “0.0 1 1”;** the stray point is a scanner artefact over a zero, and the matrix is the standard form.

Step 1: run the four tests

Test	Computation	Result	Conclusion
Diagonal	$S_{11} = S_{22} = S_{33} = S_{44} = 0$	all zero	Matched at every port
Transpose	$S_{13} = S_{31} = 1, S_{23} = S_{32} = -1,$ $S_{14} = S_{41} = 1, S_{24} = S_{42} = 1$	$[S] = [S]^T$	Reciprocal , so passive and non-ferrite
Column sums	$ 0 ^2 + 0 ^2 + 1 ^2 + 1 ^2 = 2$ for every column	$2 \neq 1$	Not unitary as printed: the $1/\sqrt{2}$ factor has been omitted
Column dots	$C_1 \cdot C_2^* = (0)(0) + (0)(0) + (1)(-1) + (1)(1) = 0;$ $C_3 \cdot C_4^* = (1)(1) + (-1)(1) + 0 + 0 = 0$	all zero	Phases are consistent; only the scale is wrong

- **Judging the condition:** divide the printed matrix by $\sqrt{2}$. Then every column sums to $\frac{1}{2} + \frac{1}{2} = 1$ and the device is **lossless**.
Say this explicitly: it is the “judge the condition” mark.

Step 2: map the zeros and read the signs

- **Zero blocks:** $S_{12} = S_{21} = 0$, so **ports 1 and 2 are isolated**; $S_{34} = S_{43} = 0$, so **ports 3 and 4 are isolated**. Two isolated pairs on a matched, lossless, reciprocal 4-port.
- **Column 3** reads $(1, -1)$ down the collinear ports: the signs **flip**, so port 3 is the **difference (E) arm**.
- **Column 4** reads $(1, 1)$: the signs **match**, so port 4 is the **sum (H) arm**.
- **Equal magnitudes** everywhere off the zero blocks, so every split is 50/50.

Answer

- The device is a magic tee (hybrid tee), with ports 1 and 2 the collinear arms, port 3 the E-arm and port 4 the H-arm. Normalised:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- Diagram: reproduce the 3D sketch of Problem 8 with all four arms labelled.

Properties, one line each

- **Matched** at all four ports, so no port reflects: $S_{ii} = 0$, return loss infinite in the ideal case.
- **Reciprocal**, $[S] = [S]^T$: passive waveguide, no ferrite, no gain.
- **Lossless** once normalised: all incident power is redistributed, none dissipated.
- **Two isolated pairs**: 1-2 and 3-4. Two sources on ports 1 and 2 cannot see each other, and the E-arm and H-arm are decoupled.
- **Equal power division**, $|S_{13}|^2 = \frac{1}{2}$: every drive splits in half, i.e. a 3 dB hybrid.
- **Phase**: E-arm drive gives antiphase outputs, H-arm drive gives in-phase outputs.
- **Uses**: power combiner and splitter, duplexer, balanced mixer, monopulse comparator, impedance bridge.

Problem 14: Identify and complete a symbolic 4-port matrix 5 8

74 Bh, 81 Bh · the same device as Problem 13, printed before the unitary step

QUESTION AS PRINTED 81 Bh [5] and 74 Bh [8]

Identify and explain, along with a neat diagram, the properties of the passive microwave device described by the following S-matrix:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

Step 1: read the structure before doing any algebra

Feature	Where	What it means
$S_{33} = S_{44} = 0$	bottom-right block	Ports 3 and 4 are matched to the junction
$S_{34} = S_{43} = 0$	bottom-right block	Ports 3 and 4 are isolated from each other
Column 3 is (S_{13}, S_{13})	signs match	Port 3 drives the collinear arms in phase : the H-arm
Column 4 is $(S_{14}, -S_{14})$	signs flip	Port 4 drives them in antiphase : the E-arm
$[S] = [S]^T$	everywhere	Reciprocal, hence passive and isotropic

- Four ports, two matched side arms, one sum arm and one difference arm: **this is a magic tee**. Note that the port labels are the reverse of Problem 13: here **port 3 is the H-arm and port 4 the E-arm**. **Read the signs, not the numbers**.

Step 2: finish the job the printed matrix leaves undone

- The matrix still carries $S_{11}, S_{12}, S_{22}, S_{13}, S_{14}$ as unknowns. Apply the unitary property exactly as in Problem 8:

Product	Equation	Result
$R_3 \cdot C_3$	$ S_{13} ^2 + S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$
$R_4 \cdot C_4$	$ S_{14} ^2 + -S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$
$R_3 \cdot C_4$	$S_{13}S_{14}^* - S_{13}S_{14}^* = 0$	satisfied identically, no new information
$R_1 \cdot C_2$	$S_{11}S_{12}^* + S_{12}S_{22}^* + S_{13} ^2 - S_{14} ^2 = 0$, and the last two cancel	$S_{11}S_{12}^* + S_{12}S_{22}^* = 0$
$R_1 \cdot C_1$	$ S_{11} ^2 + S_{12} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$ S_{11} ^2 + S_{12} ^2 = 0$, so $S_{11} = S_{12} = 0$
$R_2 \cdot C_2$	$ S_{12} ^2 + S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{22} = 0$

Answer

- Magic tee, with the completed matrix

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

which is Problem 13's matrix with columns 3 and 4 interchanged, as the port labelling demands.

- **The strongest single line in this answer:** nothing was assumed about S_{11} , S_{12} or S_{22} , yet all three came out zero. Matching the two side arms of a 4-port **forces** the collinear arms to be matched and isolated. That is precisely what a 3-port cannot do (§3.5).
- Properties and uses: as listed in Problem 13.

Problem 15: Judge the condition of a given S-matrix 6

75 Bh · the isolation is already printed in, so only the scale is left to judge

QUESTION AS PRINTED 75 Bh [6]

Which passive microwave device is explained by this S-matrix? Judge the condition and explain its characteristics. $[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$

Step 1: identify

- Four ports; $S_{33} = S_{44} = 0$ and $S_{34} = S_{43} = 0$, so ports 3 and 4 are matched and mutually isolated.
- $S_{12} = S_{21} = 0$ is **already written in**, so ports 1 and 2 are isolated too: the collinear pair.
- Column 3 is $(S_{13}, -S_{13})$, signs flipping → **E-arm**. Column 4 is (S_{14}, S_{14}) , signs matching → **H-arm**.
- $[S] = [S]^T \rightarrow$ reciprocal, so passive.
- Magic tee, with the port labelling of Problem 13.

Step 2: judge the condition (what the question actually rewards)

Test	Computation	Result	Conclusion
Diagonal	S_{11}, S_{22} left as unknowns; $S_{33} = S_{44} = 0$	side arms matched	Collinear match not yet stated
Transpose	every $S_{ij} = S_{ji}$	symmetric	Reciprocal
Column 3 sum	$ S_{13} ^2 + -S_{13} ^2 = 1$	$S_{13} = \frac{1}{\sqrt{2}}$	Equal, antiphase split from the E-arm
Column 4 sum	$ S_{14} ^2 + S_{14} ^2 = 1$	$S_{14} = \frac{1}{\sqrt{2}}$	Equal, in-phase split from the H-arm
Column 1 sum	$ S_{11} ^2 + 0 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{11} = 0$	Port 1 is matched after all
Column 2 sum	$0 + S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$	$S_{22} = 0$	Port 2 likewise

- **The condition:** the matrix describes a **lossless, reciprocal, matched-at-all-four-ports** device, and this is **consistent** only because it has four ports. The same set of demands on a 3-port is impossible (§3.5).
- Completed:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Characteristics

- **3 dB equal division:** $|S_{13}|^2 = |S_{14}|^2 = \frac{1}{2}$, so every input splits half and half.
- **Double isolation:** 1-2 and 3-4. Two transmitters on the collinear arms cannot load each other; the E-arm and H-arm cannot leak into each other.
- **Phase discrimination:** sum at the H-arm, difference at the E-arm, which is what makes it usable as a comparator.

- **Bidirectional**: being reciprocal, it combines as readily as it splits.
- **Limitation**: the ideal matrix assumes matching irises at the junction and holds over a limited bandwidth.

Problem 16: Two radar transmitters, twice the power into one antenna 6 8

73 Bh, 74 Ma · a design question answered entirely by one matrix multiplication

QUESTION AS PRINTED 73 Bh [8] and 74 Ma [6]

Suppose there are two identical radar transmitters and a few passive devices in equipment stock. A particular application requires twice more input power to an antenna than either transmitter can deliver. As an RF engineer, give your appropriate solution with necessary figures, mathematics and sufficient explanation.

Given

- Two identical transmitters, each able to deliver P watts.
- One antenna, which must receive $2P$ watts.

Step 1: choose the device, and say why

- **Choice**: a **magic tee** used as a power **combiner** (a 3 dB hybrid coupler works identically).
- **Why not a simple T-junction or Y-junction**:
 - A plain junction gives $S_{12} \neq 0$, so each transmitter drives power into the other one: mutual loading, pulling of the oscillators, and possible damage.
 - Its collinear ports are not matched ($S_{11} = S_{22} = \frac{1}{2}$ for a tee, Problem 6), so both transmitters see a high VSWR.
- The magic tee cures both: $S_{12} = 0$ (perfect isolation) and $S_{ii} = 0$ (matched at every port).

Step 2: the connection

Port	Connected to	Reason
1 (collinear)	Transmitter A	Isolated from port 2
2 (collinear)	Transmitter B	Isolated from port 1
3 (E-arm)	Matched dummy load	Takes the difference , which is zero when the drives are balanced
4 (H-arm)	Antenna	Takes the sum of the two drives

- The two feed lines must be cut to the **same electrical length**, so that the transmitters arrive in phase.

Step 3: the mathematics

- Both transmitters equal and in phase, nothing incident on the side arms:

$$[a] = [a \quad a \quad 0 \quad 0]^T, \quad |a|^2 = P$$

- Multiply $[b] = [S][a]$ with the magic-tee matrix:

$$\begin{aligned}
 b_3 &= \frac{1}{\sqrt{2}}(a_1 - a_2) = \frac{1}{\sqrt{2}}(a - a) = 0 \\
 b_4 &= \frac{1}{\sqrt{2}}(a_1 + a_2) = \frac{1}{\sqrt{2}}(2a) = \sqrt{2}a \\
 b_1 &= \frac{1}{\sqrt{2}}(a_3 + a_4) = 0, \quad b_2 = \frac{1}{\sqrt{2}}(-a_3 + a_4) = 0
 \end{aligned}$$

Step 4: read the powers

Port	Wave	Power	Meaning
4, antenna	$b_4 = \sqrt{2}a$	$ b_4 ^2 = 2 a ^2 = 2P$	The requirement, met exactly
3, dummy load	$b_3 = 0$	0	Nothing wasted when the drives are balanced
1 and 2, transmitters	$b_1 = b_2 = 0$	0	No reflection back into either transmitter
Total out		2P	Equals the total in, as a lossless network requires

Answer

- Feed the two transmitters into the collinear arms of a magic tee, antenna on the H-arm, matched load on the E-arm. The antenna then receives $2P$ watts, twice what either transmitter alone can deliver.

Explanation, which carries the remaining marks

- **Why the sum appears at the H-arm:** an H-plane junction excites the two collinear arms in phase, so by reciprocity two in-phase drives add there.
- **Why the dummy load matters:** if the transmitters are unequal or out of phase, $b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) \neq 0$ and the imbalance is **absorbed** by the load instead of being reflected into the transmitters. The load must be rated for the worst-case imbalance, i.e. the full power of one transmitter.
- **Phase error costs power:** with a phase error ϕ between the drives, the antenna receives $2P \cos^2(\phi/2)$ and the load takes the rest. A 20° error already loses 0.13 dB.
- **Alternatives:** a 3 dB branch-line or rat-race hybrid does the same job in microstrip; a Wilkinson combiner does it with an isolation resistor when the device must be planar.

Problem 17: ASR duplexer delivering half the transmitter power 8 10

73 Ma, 76 Bh · the same magic tee, driven from the other end

QUESTION AS PRINTED 76 Bh [10] and 73 Ma [8]

A radar installation engineer is given the responsibility of installing an Airport Surveillance Radar that requires half the input power to the antenna than the transmitter can deliver, using a duplexer. Prepare its S-matrix. (73 Ma: choose a suitable passive microwave device to split power in half and explain its properties.)

Given

- A transmitter delivering P watts.
- The antenna must receive $P/2$ watts, and the receiver must share the same antenna while staying isolated from the transmitter.

Step 1: choose the device

- **Choice:** a **magic tee used as a duplexer**. It is the only passive device that splits power exactly in half *and* provides the transmitter-to-receiver isolation a duplexer must have.
- A resistive splitter would also halve the power but would burn the other half and give no isolation.

Step 2: the connection

Port	Connected to	Reason
4 (H-arm)	Transmitter	Drives both collinear arms equally and in phase
1 (collinear)	Antenna	Receives half the transmitter power
2 (collinear)	Matched dummy load	Absorbs the other half, and balances the bridge
3 (E-arm)	Receiver	Isolated from the H-arm, so the transmit pulse never reaches it

Step 3: transmit path

- Drive vector $[a] = [0 \ 0 \ 0 \ a]^T$ with $|a|^2 = P$.
- From $[b] = [S][a]$:

$$b_1 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_2 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_3 = 0$$

- Powers:

$$|b_1|^2 = \frac{1}{2}|a|^2 = \frac{P}{2}, \quad |b_2|^2 = \frac{P}{2}, \quad |b_3|^2 = 0$$

- **The antenna receives exactly $P/2$** , which is the requirement, and the receiver on the E-arm receives **nothing** from the transmitter, which is the point of a duplexer.

Step 4: receive path

- The echo arrives at port 1 alone: $[a] = [a_1 \ 0 \ 0 \ 0]^T$ with $|a_1|^2 = P_{echo}$.
- $b_3 = \frac{1}{\sqrt{2}}a_1$ and $b_4 = \frac{1}{\sqrt{2}}a_1$, so

$$|b_3|^2 = \frac{P_{echo}}{2} \quad \text{to the receiver,} \quad |b_4|^2 = \frac{P_{echo}}{2} \quad \text{back into the transmitter}$$

- **Nothing reaches port 2**, since $S_{21} = 0$.

Answer

- The duplexer is the magic tee, whose S-matrix is

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

with port 4 the transmitter, port 1 the antenna, port 2 the matched load and port 3 the receiver.

Properties and the honest limitation

- **Equal split:** $|S_{14}|^2 = |S_{24}|^2 = \frac{1}{2}$, a 3 dB hybrid.
- **Isolation:** transmitter (port 4) to receiver (port 3) is $S_{34} = 0$, so the receiver front end is protected from the full transmit pulse.
- **Matched at every port**, so neither the transmitter nor the antenna sees a standing wave.
- **The 3 dB penalty is real and should be stated:** half the transmit power goes into the dummy load, and half the echo goes back to the transmitter. The magic-tee duplexer therefore costs 3 dB in each direction.
- **Better in practice:** a **ferrite circulator** duplexer, $[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, routes the *whole* transmit power to the antenna and the *whole* echo to the receiver, with no 3 dB loss. Quote it as the improvement whenever the question does not explicitly demand the half-power split, as **76 Bh** does.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.